exploratory-rmbl

Linda Herrera

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(dplyr)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)
library(ggridges)
```

Data

```
# Trail Creek Isotope
tc_isotope <- read_csv(here("data", "TC-isotopes_april-june.csv"), show_col_types = FALSE) |
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stdev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stdev) |>
```

```
  mutate(site = case_when( # creating a new column of the site
    str_detect(sample, "UTC") ~ "utc",
    str_detect(sample, "TC")  ~ "tc",
    TRUE ~ NA_character_ )) |> # For names without UTC or TC
  filter(!is.na(site)) |>
  mutate(date = mdy(date)) # telling r the date format

# Downstream Trail Creek
tc_data <- read_csv(here("data", "TrailCreek_20260415_export20260630.csv"), show_col_types =
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
         dp = differential_pressure_k_pa,
         ap = absolute_pressure_k_pa,
         temp = temperature_c,
         water_level = water_level_m,
         bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
         date_only = as_date(date_time_parsed)) |> # making a date column
  group_by(date_only) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
            avg_ap = mean(ap, na.rm = TRUE),
            avg_temp = mean(temp, na.rm = TRUE),
            avg_water_level = mean(water_level, na.rm = TRUE),
            avg_bp = mean(bp, na.rm = TRUE),
            total_readings = n(), # the amount of data it used to average
            .groups = "drop") |>
  mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_data <- read_csv(here("data", "UTC_20260403_export20260630.csv"), show_col_types = FALSE
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt,  # renaming columns to be shorter
         dp = differential_pressure_k_pa,
         ap = absolute_pressure_k_pa,
         temp = temperature_c,
         water_level = water_level_m,
         bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
         date_only = as_date(date_time_parsed)) |> # making a date column
  group_by(date_only) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE),  # averaging the data based on the day
            avg_ap = mean(ap, na.rm = TRUE),
```

```r
            avg_temp = mean(temp, na.rm = TRUE),
            avg_water_level = mean(water_level, na.rm = TRUE),
            avg_bp = mean(bp, na.rm = TRUE),
            total_readings = n(), # the amount of data it used to average
            .groups = "drop") |>
  mutate(site = "utc") # creating a new column of the site
```